

Asyam Varshney

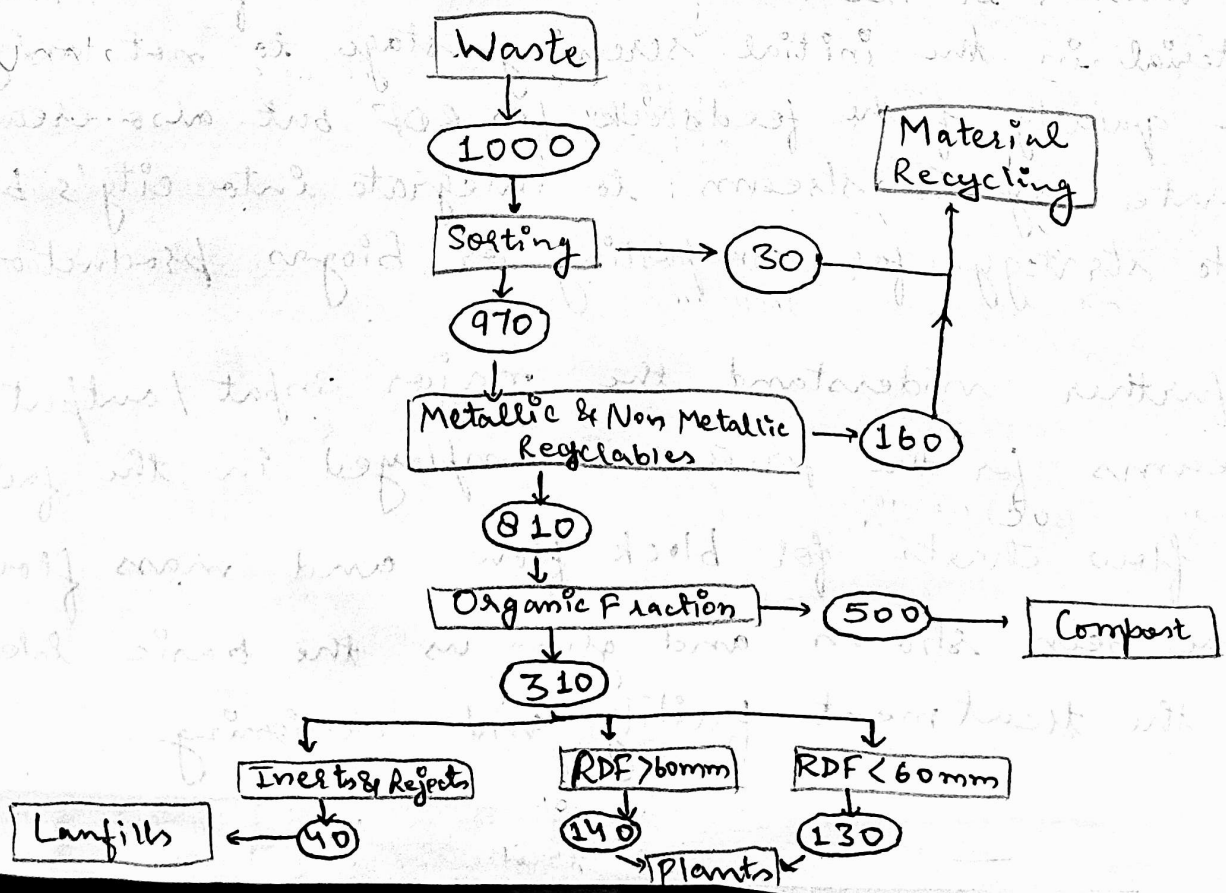
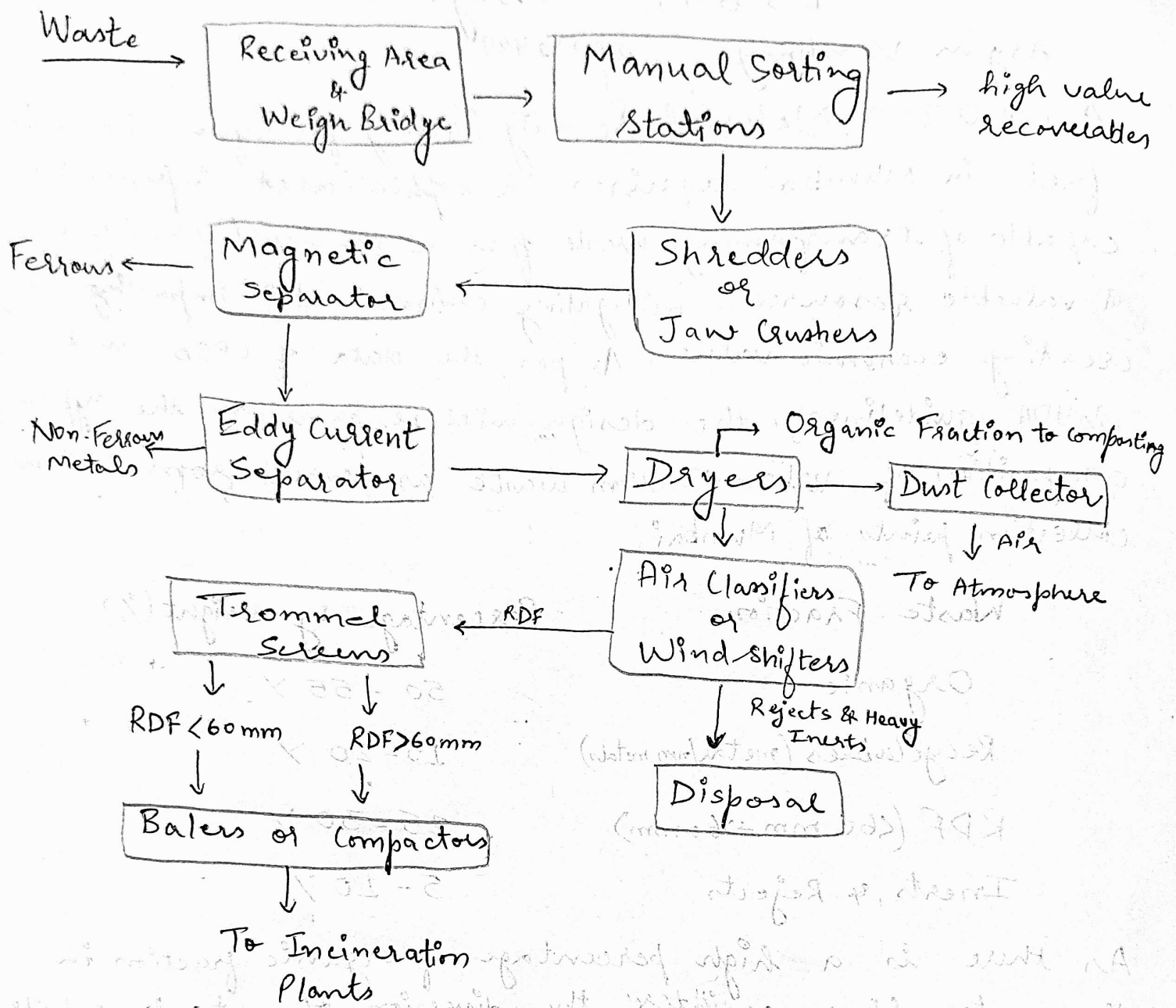
25M1429

A 1000 TPD Material Recovery Facility for Refuse - Derived fuel in Mumbai requires a sophisticated infrastructure capable of transforming waste from a municipal liability into a valuable resource, mitigating environmental impacts and creating economic value. As per the data of CPCB and MoHUA guidelines, the design will be based on the typical composition of Urban Indian waste as sourced from various collection points of Mumbai.

Waste Fraction	Percentage by Weight (%)
Organic	50 - 55 %
Recyclables (metal/non-metals)	15 - 20 %
RDF (<60 mm + >60 mm)	25 - 30 %
Inerts & Rejects	5 - 10 %

As there is a high percentage of organic fraction in the waste, it necessitates the diversion of wet, low calorific material in the initial screening stage to not only improve the quality of the feedstock for RDF but also creates a separate organic stream to integrate into city's broader waste strategy for composting or biogas production.

To further understand the major input/output streams for the processes employed in the facility the flow charts for block flow and mass flow have been shown and give us the basic idea of the treatment facility and designing.



- Fines Rich RDF (<60mm) : A high calorific fraction suitable for fluidized bed combustion or as a supplementary fuel in specific industrial processes.
- Refuse Derived fuel (>60mm) : Primary RDF product, densified into bales / chips for efficient transport and use in waste to energy incineration plants and cement kilns.
- Recovered Organic Fraction : Biodegradable materials separated and routed ~~to~~ for composting, or other biological treatment.
- Recyclables : - Metallic & Non metallic can be sold as commodities in secondary material markets.
- Inerts & Rejects : ~~The~~ Fraction with no recoverable value and sent to sanitary landfills.

The proposed design integrates a combination of mechanical, automated and manual systems specifically chosen for suitability in handling & efficiency for handling this type of waste composition. The facilities operational energy footprint based on Min. of New and Renewable Energy is projected to be 100 - 120 kWh per tonne of processed waste. In facility daily and hourly material balance break down is shown below assuming 16 hr workday.

Stream	Daily (tonnes)	hourly (tonnes)
RDF < 60mm	140	8.75
RDF > 60mm	130	8.125
Organic Fraction	500	31.25
Material Recycled	190	11.875
Inerts & Rejects	40	2.5

$$\begin{aligned}
 \text{Energy Consumed} &= 1000 \text{ tonnes/day} \times 110 \text{ kWh} \\
 &= 110,000 \text{ kWh/day}
 \end{aligned}$$

Equipment	Selected Model/Type	Capacity (TPH)	Assumption/Justification
Weighbridge	Amey India 60T	60 T./load	Standard for truck entry
Trommel	60 mm dual drum	25 TPH	Size based separation
Smredder	UNTHA RS50	20 TPH	For Reducing size of waste for processing
Dryer	Rotary Drum	15 TPH	RDF < 60 mm Moisture Reduction
Magnetic Separator	Overband type	10 TPH	Ferrous Recovery
Eddy Current Separator	Steinerted	5 TPH	Non Ferrous Recovery
Manual Sorting line	10 station ^{station}	20 TPH	Flexible for Indian waste composition
Wind shifter	BRT Hutterws	10 TPH	RDF refinement
Air Classifier	Dual Stage	15 TPH	Light Heavy Separation
Baler	Horizontal autotic	10 TPH	RDF packaging
Compactor	Hydraulic twin Ram	5 TPH	Volume Reduction

An efficient facility layout is crucial for optimising material flow ensuring operational safety and maximising throughput. A well designed/planned site provides a seamless transition, preventing bottlenecks and facilitating safe movement of heavy machinery. Assuming truck turning radius of 12m, sorting line length 30m and other buffer zones & safety margins included the total land footprint will be approximates as.

Receiving Area $\rightarrow 1000 \text{ m}^2$
 Sorting & processing $\rightarrow 3000 \text{ m}^2$
 Bale storage $\rightarrow 1500 \text{ m}^2$
 Truck movement & parking $\rightarrow 2000 \text{ m}^2$
 Admin & Utilities $\rightarrow 500 \text{ m}^2$
 Refuse Storage $\rightarrow 400 \text{ m}^2$

Total land required = 8400 m^2
 $\approx 2 - 2.5$ acres land

The distinct functional zones in the facility -

→ Processing Area - core of the facility, housing the main building where integrated machinery lines for sorting, shredding, drying and baling are installed. The design will prioritize linear flow to minimize material handling complexity.

→ Storage & Logistics :- This area functions as a critical operational buffer decoupling the continuous 24/7 16 hr processing line from variable logistics of RDF dispatch and recyclable collection.

→ Supporting Infrastructure - Space allocated for administration office, weighbridge, control room, utilities and other essential staff facilities.

Based on various literature & vendor quotes.

Components

Costs (in Crore ₹)

Equipments &
Machinery

350

Civil works

100

Installation & Setup

30

Land (considering
cheaper lands)

30-35

Labour & training

2

Contingency

3-4

Total cost

=

~~₹ 515~~ 515 - 525 Crore

This cost structure underscores the project sensitivity to land acquisition timelines and international equipment markets. They must be weighed against the substantial environmental and social benefits & returns on this critical urban infrastructure investments. This outlines a technically sound, economically projected and environmentally sustainable solution for Mumbai's MSW challenges.